

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSE
Course Outcome	2

Case Study Topic: Apache Cassandra LSM-Tree – Compaction Strategy and Write Amplification

Word Done Summary:

In this case study, we analyzed the **Log-Structured Merge Tree (LSM-Tree)** architecture used in Apache Cassandra and other modern write-optimized databases such as RocksDB, ScyllaDB, and LevelDB.

Unlike traditional B+ Trees that update data in place, LSM-Trees use an append-only approach. Incoming writes are first stored in an in-memory structure called a **MemTable**. Once the MemTable becomes full, it is flushed to disk as an immutable **SSTable**.

Over time, multiple SSTables accumulate. To improve query performance and remove obsolete records, Cassandra periodically performs **compaction**, where multiple SSTables are merged into larger SSTables.

Implementation Details:

1. LSM-Tree Architecture
2. B+ Tree Architecture
3. B+ Tree Architecture

Result:

Given Configuration :

MemTable = 1 GB

Level-0 SSTables = 8

Each SSTable = 1 GB

Level-1 Capacity = 10 GB

Compaction Activity :

Data Read = 8 GB

Data Written = 8 GB

Total Physical Writes :

16 GB

Logical Data Written :

8 GB

Write Amplification :

2×

Output:

```
Logical Writes: 8.0 GB  
Physical Writes: 16.0 GB  
Write Amplification: 2.0x
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GitHub Link:

<https://github.com/karthiktalla-9392/DSA-CASE-STUDIES>

Student Signature	Faculty Signature
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